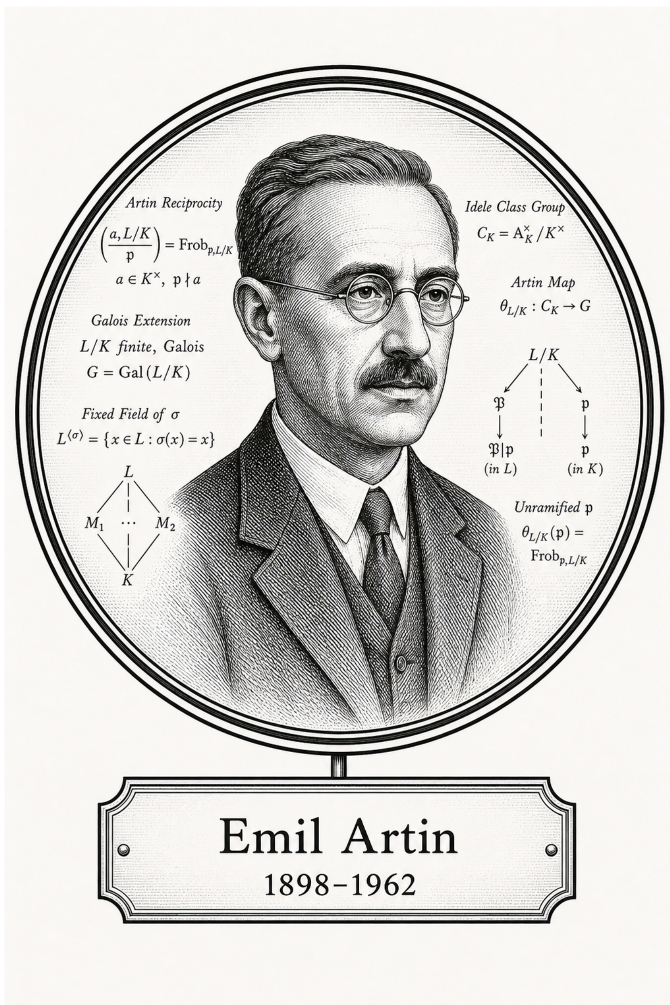
From Cantor to Ito

Algebra

Algebra



Emil Artin

1898-1962

Self-Study Notes

William Sollers

July 1, 2026

For every reader learning to make mathematics their own.

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Chapter 1

Introduction to Algebraic Structures

SERIES
From Cantor to Ito

VOLUME VI
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CHAPTER 1
Introduction to Algebraic Structures

[

Where You Are in the Journey]

Propositional Logic $\rightarrow$ Predicate Calculus $\rightarrow$ Sets & Functions $\rightarrow$ Proof Techniques $\rightarrow$ Real Analysis $\rightarrow$ **Intro to Algebraic Structures** $\rightarrow$ Linear Algebra $\rightarrow$ Topology $\rightarrow$ $\cdots$

This chapter fixes the vocabulary of algebraic structure first, then builds the standard one-law and two-law structure tower by adding one axiom at a time.

1.1 What Is a Mathematical Structure?

1.1.1 Algebraic Structures

Remark (Geometric picture). Before any specific algebra, fix what kind of thing we are studying: a set of elements together with stipulated ways to combine them, compare them, and name some of them. The axioms say which stipulations we keep. A group, a ring, and a field are the same general idea with different stipulations.

Definition (Algebraic Structure)

Definition 1.1.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Remark (Standard quantified statement)*.*

$$\text{AlgStructure}_{\mathcal{L}}(\mathcal{A}) \iff \mathcal{A} = (A, I),\ A \neq \varnothing,\ \text{and } I \text{ interprets every symbol of } \mathcal{L} \text{ on } A.$$

Remark (Interpretation)*.* The signature is the type; the structure is the instance. "Group" is a signature with a binary operation, an identity constant, and an inverse operation; a particular group interprets these on a carrier.

Remark (Bourbaki anchor)*.* Species of structure, *Theory of Sets*, Chapter IV, Section 1.

Remark (Dependencies)*.*

- Carrier Set
- First-Order Signature

Definition (First-Order Signature)

Definition 1.1.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\text{Const}_{\mathcal{L}}, \text{Func}_{\mathcal{L}}, \text{Rel}_{\mathcal{L}}, \text{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Remark (Dependencies)*.*

- Nullary Operation
- Unary Operation
- Binary Operation

Definition ($\mathcal{L}$-Structure)

Definition 1.1.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

Remark (Dependencies)*.*

- First-Order Signature
- Carrier Set

1.1.2 Laws of Composition

Definition (Binary Operation / Law of Composition)

Definition 1.1.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Remark (Standard quantified statement).

$$\mathrm{BinOp}(\star, A) \iff \forall a, b \in A,\ a \star b \in A.$$

Remark (Interpretation). The codomain A is closure: combining two elements never leaves A.

Remark (Bourbaki anchor). "Law of composition," *Algebra I*, Section 1, Definition 1. "Binary operation" is the model-theoretic alias used in this volume.

Remark (Dependencies).

- Function (Volume I recall)
- Carrier Set

Definition (n-ary Operation)

Definition 1.1.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Remark (Standard quantified statement).

$$\mathrm{Op}_n(f, A) \iff f : A^n \to A.$$

Definition (Unary Operation)

Definition 1.1.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition (Nullary Operation / Constant)

Definition 1.1.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

Remark (Forward reference: external laws). An external law or action $\Omega \times A \to A$, whose elements of Ω are operators, is introduced with actions when modules and vector spaces are reached.

1.1.3 Distinguished Elements

Remark (Geometric picture). Some elements have a privileged role under a law: one that does nothing and one that swallows everything. Defining the one-sided versions first lets the two-sided collapse become a theorem rather than an assumption.

Definition 1.1.8 (Left Identity). An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 1.1.9 (Right Identity). An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 1.1.10 (Identity / Neutral Element). An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Remark (Standard quantified statement).

$$\text{Identity}(e, \star, A) \iff e \in A \text{ and } \forall a \in A,\ e \star a = a \star e = a.$$

Proposition 1.1.11 (Identity Collapse and Uniqueness). *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.* *Go to proof.*

Remark (Proof structure). Evaluate $e_L \star e_R$ two ways.

Definition 1.1.12 (Left Absorbing Element). An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 1.1.13 (Right Absorbing Element). An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 1.1.14 (Absorbing Element). An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Remark (Standard quantified statement).

$$\text{Absorbing}(z, \star, A) \iff z \in A \text{ and } \forall a \in A,\ z \star a = a \star z = z.$$

Proposition 1.1.15 (Uniqueness of the Absorbing Element). *A two-sided absorbing element, if it exists, is unique.* *Go to proof.*

1.1.4 Inverses

Remark (Geometric picture). An inverse undoes an element, returning the identity. Undoing from the left and undoing from the right can a priori differ; associativity forces them to agree.

Definition 1.1.16 (Left Inverse). Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 1.1.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 1.1.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Remark (Standard quantified statement)*.*

$$\mathrm{Inverse}(b, a, \star, e) \iff a \star b = b \star a = e.$$

Definition 1.1.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Remark (Standard quantified statement)*.*

$$\mathrm{Invertible}(a) \iff \exists b,\ a \star b = b \star a = e.$$

Proposition 1.1.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.* *Go to proof.*

Remark (Proof structure)*.* Compute $b \star a \star c$ two ways; associativity is essential.

Definition 1.1.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 1.1.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 1.1.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.* *Go to proof.*

1.1.5 Properties of Laws

Definition 1.1.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Remark (Standard quantified statement)*.*

$$\mathrm{Assoc}(\star, A) \iff \forall a, b, c \in A,\ (a \star b) \star c = a \star (b \star c).$$

Theorem 1.1.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.* *Go to proof.*

Remark (Proof structure)*.* Induct on n, reducing every bracketing to the right-nested one.

Definition 1.1.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 1.1.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a\star b = b\star a$ for all $a, b \in A$.

Definition 1.1.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 1.1.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 1.1.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Remark (Interpretation). Distributivity is the bridge between the two laws of a ring.

Definition 1.1.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

1.1.6 Regularity Properties

Remark (Geometric picture). A cancellable element is one you can divide out of an equation even when no inverse exists. Where cancellation fails, the witnesses are zero divisors: the obstruction the ring tower is built to remove.

Definition 1.1.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \implies x = y$$

for all $x, y \in A$.

Definition 1.1.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \implies x = y$$

for all $x, y \in A$.

Definition 1.1.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 1.1.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.* *Go to proof.*

Remark (Proof structure). Left-multiply $a \star x = a \star y$ by a^{-1} and use associativity.

Definition 1.1.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

Remark (Standard quantified statement).

$$\mathrm{ZeroDivisor}(a, R) \iff a \neq 0 \text{ and } \exists b \neq 0,\ a \cdot b = 0 \text{ or } b \cdot a = 0.$$

Remark (Carried fact). In any ring, 0 is absorbing for multiplication. This is Proposition 1.1.53, derived from distributivity rather than assumed as an axiom.

1.1.7 One Internal Law

Remark (Structure tower rule). Each structure below is the previous structure plus one axiom. Closure is absorbed into the phrase "binary operation."

Definition 1.1.37 (Magma). A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 1.1.38 (Semigroup). A **semigroup** is a magma whose operation is associative.

Definition 1.1.39 (Monoid). A **monoid** is a semigroup with an identity element.

Definition 1.1.40 (Group). A **group** is a monoid in which every element has an inverse.

Definition 1.1.41 (Abelian Group). An **abelian group** is a group whose operation is commutative.

Definition 1.1.42 (Order of a Group). The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 1.1.43 (Uniqueness of the Identity in a Group). *The identity of a group is unique.* *Go to proof.*

Remark (Dependency). This specializes Proposition 1.1.11.

Proposition 1.1.44 (Uniqueness of Inverses in a Group). *Each element of a group has a unique inverse.* *Go to proof.*

Remark (Dependency). This specializes Proposition 1.1.20.

Proposition 1.1.45 (Cancellation Laws in a Group). *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 1.1.46 (Socks–Shoes). *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

1.1.8 Two Internal Laws

Remark (Convention). Throughout this volume, **ring** means unital ring. A non-unital object with ring-like addition, multiplication, associativity, and distributivity is called a **rng**.

Definition 1.1.47 (Rng). A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 1.1.48 (Ring). A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 1.1.49 (Commutative Ring). A **commutative ring** is a ring whose multiplication is commutative.

Definition 1.1.50 (Integral Domain). An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 1.1.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 1.1.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 1.1.53 (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a. Go to proof.*

Remark (Proof structure)*.* Distribute $a(0+0)$ and cancel additively in $(R, +)$.

Proposition 1.1.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. Go to proof.

Proposition 1.1.55 (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$. Go to proof.*

Proposition 1.1.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain. Go to proof.*

Proposition 1.1.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field. Go to proof.*

1.1.9 The Structural Lattice

Magma	Rng
$\Downarrow$ add associativity	$\Downarrow$ add unity
Semigroup	Ring
$\Downarrow$ add identity	$\Downarrow$ add multiplicative commutativity
Monoid	Commutative Ring
$\Downarrow$ add inverses	$\Downarrow$ add $0 \neq 1$ and no zero divisors
Group	Integral Domain
$\Downarrow$ add commutativity	$\Downarrow$ finite, or add inverses directly
Abelian Group	Field

Remark (Proof-stub inventory)*.* The one-proof-per-file pipeline for this chapter contains: identity-collapse, absorbing-unique, inverse-collapse, units-form-group, general-associativity, invertible-implies-cancellable, group-identity-unique, group-inverse-unique, group-cancellation, socks-shoes, ring-mult-zero, ring-mult-neg, domain-cancellation, field-is-domain, and finite-domain-is-field.

1.2 Relations

1.3 Operations

1.4 Laws of Operations

1.5 Functions

1.6 Distinguished Elements

1.7 Algebraic Structures

1.8 Groups

1.9 Rings

1.10 Fields

1.11 Number Systems as Structures

1.12 Number Systems as Models

Proofs

Remark (Return). Return to Theorem

Proposition (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.*

Professional Standard Proof.
TODO: professional standard proof for identity-collapse. □

Detailed Learning Proof.
TODO: detailed learning proof for identity-collapse. □

Remark (Proof structure)*.*

Remark (Dependencies)*.*

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique.*

Professional Standard Proof.
TODO: professional standard proof for absorbing-unique. ☐

Detailed Learning Proof.
TODO: detailed learning proof for absorbing-unique. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.*

Professional Standard Proof.
TODO: professional standard proof for inverse-collapse. □

Detailed Learning Proof.
TODO: detailed learning proof for inverse-collapse. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Units of a Monoid Form a Group). *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.*

Professional Standard Proof.
TODO: professional standard proof for units-form-group. □

Detailed Learning Proof.
TODO: detailed learning proof for units-form-group. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.*

Professional Standard Proof.
TODO: professional standard proof for general-associativity. ☐

Detailed Learning Proof.
TODO: detailed learning proof for general-associativity. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.*

Professional Standard Proof.
TODO: professional standard proof for invertible-implies-cancellable. ☐

Detailed Learning Proof.
TODO: detailed learning proof for invertible-implies-cancellable. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-identity-unique. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-identity-unique. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.*

Professional Standard Proof.
TODO: professional standard proof for group-inverse-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for group-inverse-unique. □

Remark (Proof structure)*.*

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for group-cancellation. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Socks–Shoes)**.** *In a group,* $(ab)^{-1} = b^{-1}a^{-1}$.

Professional Standard Proof.
TODO: professional standard proof for socks-shoes. □

Detailed Learning Proof.
TODO: detailed learning proof for socks-shoes. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Zero). *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a.*

Professional Standard Proof.
TODO: professional standard proof for ring-mult-zero. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-zero. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Additive Inverse). *In a ring, $a \cdot (-b) = -(a \cdot b) = (-a) \cdot b$. In a ring, $(-1) \cdot a = -a$.*

Professional Standard Proof.
TODO: professional standard proof for ring-mult-neg. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-neg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for domain-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for domain-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain)**.** *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for field-is-domain. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field.*

Professional Standard Proof.
TODO: professional standard proof for finite-domain-is-field. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-domain-is-field. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws in a Group). *Let G be a group and let $a, b, c \in G$. If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for group-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity)**.** *In a group, the identity element is unique.*

Professional Standard Proof.
TODO: professional standard proof for identity-element-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for identity-element-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses)**.** *In a group, every element has a unique inverse.*

Professional Standard Proof.
TODO: professional standard proof for inverse-element-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for inverse-element-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains). *Let R be an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for abstract-domain-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for abstract-domain-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain). *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for abstract-field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for abstract-field-is-domain. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws)**.** *Let G be a group and let $a, b, c \in G$. Then:*

(i) ***Left cancellation:*** $ab = ac \implies b = c$.

(ii) ***Right cancellation:*** $ba = ca \implies b = c$.

Professional Standard Proof.
TODO: professional standard proof for group-cancellation-in-groups. □

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation-in-groups. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity). *Let G be a group. The identity element of G is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-identity-unique. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-identity-unique. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses). *Let G be a group. For each $a \in G$, the inverse of a is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-inverse-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for group-inverse-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Socks-Shoes Property)**.** *Let G be a group and let $a, b \in G$. Then*

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Professional Standard Proof.
TODO: professional standard proof for group-socks-shoes. □

Detailed Learning Proof.
TODO: detailed learning proof for group-socks-shoes. □

Remark (Proof structure)*.*

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Commutativity from Idempotence). *In any Boolean ring R,*

$$pq = qp \quad \textit{for all } p, q \in R.$$

Hence every Boolean ring is a commutative ring.

Professional Standard Proof.
The proof of Proposition **??** established

$$pq + qp = 0 \qquad \text{for all } p, q \in R.$$

Hence $qp = -(pq)$. By Proposition **??**, applied to the element pq, we have $-(pq) = pq$. Therefore $qp = pq$. □

Detailed Learning Proof.
The characteristic-2 proof produced a stronger intermediate identity: $pq + qp = 0$. This says that qp is the additive inverse of pq. But in a Boolean ring every element is its own negative, so the additive inverse of pq is just pq itself. Therefore $qp = pq$. □

Remark (Proof structure). Reuse the intermediate identity from the characteristic-2 proof, then replace additive inverse by self-negation.

Remark (Dependencies).

- Self Additive Inverse / Characteristic 2.
- Self Negation.
- Commutative Ring.

Remark (Return). Return to Theorem

Corollary (Order is a Power of Two). *A finite Boolean ring has order 2^n for some $n \geq 0$. In particular, there is no Boolean ring of order 3.*

Professional Standard Proof.
By the finite case of Stone representation (Theorem **??**), a finite Boolean ring is isomorphic to $\mathcal{P}(X)$ for some finite set X. If $|X| = n$, then $|\mathcal{P}(X)| = 2^n$. Since 3 is not a power of 2, no Boolean ring has order 3. □

Detailed Learning Proof.
Finite Boolean rings are set rings in disguise. Stone representation says that every finite one looks like the power set of a finite set. A set with n elements has exactly 2^n subsets, so the ring has 2^n elements. The number 3 is not on that list. □

Remark (Proof structure). Invoke the finite Stone representation and count subsets.

Remark (Dependencies).

- Stone Representation for Boolean Rings.
- Cardinality of a finite power set.

Remark (Return). Return to Theorem

Proposition (Self Additive Inverse / Characteristic 2)**.** *In any Boolean ring R,*

$$p + p = 0 \quad \text{for every } p \in R.$$

Professional Standard Proof.
Let $p, q \in R$. By B1 the element $p + q$ is idempotent, so

$$(p + q)(p + q) = p + q. \tag{$*$}$$

Expand the left side by distributivity (R3), applied on both sides:

$$(p + q)(p + q) = p(p + q) + q(p + q) = pp + pq + qp + qq.$$

Apply B1 to $pp = p$ and $qq = q$, and regroup by additive associativity (R1):

$$p + q = p + pq + qp + q.$$

Cancel p and q using additive cancellation (Proposition 1.1.45):

$$pq + qp = 0 \qquad \text{for all } p, q \in R. \tag{$**$}$$

Specialize $(**)$ to $q = p$ and apply B1 once more:

$$p + p = pp + pp = 0.$$

□

Detailed Learning Proof.
The only new fact over a plain ring is that everything squares to itself. Apply it to a sum, because squaring a sum is where addition and multiplication are forced to interact.

Square $p+q$ two ways: it equals itself by idempotence, and it expands to the four products pp, pq, qp, qq by distributivity. There is no $2pq$ shortcut because multiplication is not assumed commutative.

Idempotence collapses pp to p and qq to q, leaving $p + pq + qp + q$. Match this against $p + q$ and cancel the shared p and q: this leaves $pq + qp = 0$.

Put $q = p$. Idempotence turns $pp + pp$ into $p + p$, so $p + p = 0$. Characteristic 2 was never assumed; it is forced by idempotence. □

Remark (Proof structure). Use B1 on $(p + q)^2$, expand by R3, cancel the outer terms by R1, and specialize the resulting identity to $q = p$.

Remark (Dependencies).

- Boolean Ring (B1).
- Ring axioms R1 and R3.
- Additive cancellation.
- Lean reference: `BooleanAlgebraDefinitions.lean`.

Remark (Return). Return to Theorem

Proposition (Self Negation)**.** *In any Boolean ring R,*

$$-p = p \quad \textit{for every } p \in R.$$

Professional Standard Proof.
By Proposition **??**, $p + p = 0$. By the additive-inverse clause of R1, $p + (-p) = 0$. Both p and $-p$ are therefore additive inverses of p; by uniqueness of additive inverses (Proposition 1.1.44), $-p = p$. □

Detailed Learning Proof.
An additive inverse of p is anything that can be added to p to reach 0. The ring gives one such element, $-p$. Proposition **??** hands us a second: p itself, since $p + p = 0$. Inverses in an abelian group are unique, so the two candidates are the same element: $-p = p$. The minus sign is therefore pure decoration in a Boolean ring. □

Remark (Proof structure)*.* Use characteristic 2 to make p an additive inverse of itself, then use uniqueness of additive inverses.

Remark (Dependencies).

- Self Additive Inverse / Characteristic 2.
- Ring axiom R1.
- Uniqueness of additive inverses.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains)**.** *Let R be an integral domain and $a, b, c \in R$ with $a \neq 0$. Then*

$$ab = ac \implies b = c.$$

Professional Standard Proof.
TODO: professional standard proof for domain-cancellation. ☐

Detailed Learning Proof.
TODO: detailed learning proof for domain-cancellation. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Additive Inverse). *Let R be a ring. For all $a, b \in R$,*

$$a \cdot (-b) = -(a \cdot b) \quad \textit{and} \quad (-a) \cdot b = -(a \cdot b).$$

In any ring with unity, $(-1) \cdot a = -a$.

Professional Standard Proof.
TODO: professional standard proof for ring-mult-neg. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-neg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Zero). *Let R be a ring. For all $a \in R$,*

$$a \cdot 0 = 0 \cdot a = 0.$$

Professional Standard Proof.
TODO: professional standard proof for ring-mult-zero. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-zero. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Stone Representation for Boolean Rings). *Every Boolean ring is isomorphic to a field of sets. Every finite Boolean ring is isomorphic to $\mathcal{P}(X)$ for some finite set X, hence has $2^{|X|}$ elements.*

Professional Standard Proof.
TODO: professional standard proof for stone-representation-boolean-rings. ☐

Detailed Learning Proof.
TODO: detailed learning proof for stone-representation-boolean-rings. ☐

Remark (Proof structure).

Remark (Dependencies).

- Boolean Ring.
- TODO: prime ideals or ultrafilters.

Remark (Return). Return to Theorem

Proposition (Characteristic of a Field)**.** *Let* $\mathbb{F}$ *be a field. The* characteristic *of* $\mathbb{F}$ *is the smallest positive integer* n *such that*

$$\underbrace{1 + 1 + \cdots + 1}_{n} = 0,$$

or 0 *if no such* n *exists. The characteristic of a field is either* 0 *or a prime* p*.*

Professional Standard Proof.
TODO: professional standard proof for field-characteristic. □

Detailed Learning Proof.
TODO: detailed learning proof for field-characteristic. □

Remark (Proof structure)*.*

Remark (Dependencies)*.*

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Nonzero Scalars Have Inverses)**.** *Let $\mathbb{F}$ be a field and $a \in \mathbb{F}$ with $a \neq 0$. Then there exists a unique $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$.*

Professional Standard Proof.
TODO: professional standard proof for field-inverse-exists. □

Detailed Learning Proof.
TODO: detailed learning proof for field-inverse-exists. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain)**.** *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for field-is-domain. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Zero Product Property in a Field)**.** *Let $\mathbb{F}$ be a field and $a, b \in \mathbb{F}$. Then*

$$ab = 0 \implies a = 0 \ \text{ or } \ b = 0.$$

Professional Standard Proof.
TODO: professional standard proof for field-zero-product. ☐

Detailed Learning Proof.
TODO: detailed learning proof for field-zero-product. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

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Chapter 2

Abstract Algebra

2.1 Induction

2.1.1 Induction

Definitions and Theorems

2.2 Integers

2.2.1 Integers

Basic Definitions and Theorems

2.3 Modular Arithmetic

2.3.1 Modular Arithmetic

Basic Definitions and Theorems

Proofs

Capstone

Chapter 3

Algebraic Geometry

SERIES	VOLUME VI	BOOK I	CHAPTER 3
From Cantor to Ito	Algebra	Algebra	Algebraic Geometry

3.1 Algebraic Geometry

3.1.1 Preliminary Definitions

Definition 3.1.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Remark (Standard quantified statement)*.* A ring is a set equipped with addition and multiplication satisfying the additive-group, multiplicative-associativity, multiplicative-identity, and distributive laws.

Remark (Interpretation)*.* A ring generalizes a field by dropping the requirement that nonzero elements have multiplicative inverses, and by not requiring multiplication to be commutative. When multiplication is commutative, we call R a *commutative ring*.

Definition 3.1.2 (Commutative Ring)**.** A ring R is *commutative* if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Remark (Standard quantified statement)*.* A ring R is commutative when multiplication in R is commutative.

Remark (Structural Summary)*.* The hierarchy of algebraic structures encountered so far is:

Structure	Additive Group	Multiplicative Structure
Ring	Abelian	Associative, with identity
Commutative Ring	Abelian	Associative, commutative, with identity
Field	Abelian	Abelian on nonzero elements

Every field is a commutative ring, but not every commutative ring is a field.

Polynomial Rings

Definition 3.1.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Remark (Standard quantified statement)*.* A polynomial over R is a finite formal sum $\sum_i a_i x^i$ with coefficients $a_i \in R$.

Remark (Interpretation)*.* The indeterminate x is not a variable ranging over values in R. It is a formal placeholder that encodes the coefficient sequence. Two polynomials are equal if and only if all their coefficients are equal.

Definition 3.1.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is

$$\deg(f) := n.$$

The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Remark (Standard quantified statement)*.* For a nonzero polynomial $f = a_n x^n + \cdots + a_0$ with leading coefficient $a_n \neq 0$, the degree of f is n.

Remark (Interpretation)*.* The zero polynomial $f = 0$ has no leading coefficient. Its degree is left undefined, or assigned $-\infty$ by convention to preserve the identity $\deg(fg) = \deg(f) + \deg(g)$.

Definition 3.1.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:

$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$

$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Remark (Standard quantified statement)*.* The set $R[x]$ is the collection of finite formal sums in x with coefficients in R, equipped with coefficientwise addition and Cauchy-product multiplication.

Example (Polynomial arithmetic over the integers)*.* Let $R = \mathbb{Z}$ and consider

$$f = 2x^2 + 3x + 1, \qquad g = x + 4 \quad \in \mathbb{Z}[x].$$

Then:

$$f + g = 2x^2 + 4x + 5,$$
$$f \cdot g = 2x^3 + 11x^2 + 13x + 4.$$

Remark (Structural Summary)*.* With these operations, $R[x]$ is itself a commutative ring. The original ring R embeds into $R[x]$ as the constant polynomials.

Property	Holds in $R[x]$?
Commutative ring	Always
Field	Only in degenerate cases
R embeds in $R[x]$	Always

Polynomial Rings in Several Variables

Definition 3.1.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \dots, x_n],$$

is defined inductively by

$$R[x_1, \dots, x_n] := R[x_1, \dots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_\alpha a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Remark (Standard quantified statement). The ring $R[x_1, \ldots, x_n]$ consists of finite formal sums $\sum_\alpha a_\alpha x^\alpha$ indexed by multi-indices.

Definition 3.1.7 (Monomial). A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Remark (Standard quantified statement). A monomial in $R[x_1, \ldots, x_n]$ is a single multi-index power x^α.

Definition 3.1.8 (Total Degree). The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Remark (Standard quantified statement). The total degree of x^α is $|\alpha| = \alpha_1 + \cdots + \alpha_n$.

Example (Example). In $\mathbb{R}[x, y, z]$, the polynomial

$$f = 3x^2y + xy^2z - 5z^3$$

has three terms with total degrees 3, 4, and 3 respectively. Hence $\deg(f) = 4$.

Remark (Relevance to Algebraic Geometry). Polynomial rings in several variables are the foundational algebraic object of algebraic geometry. The geometric objects studied in subsequent sections — affine varieties, ideals, coordinate rings — are all defined in terms of $k[x_1, \ldots, x_n]$ for a field k. The interplay between the algebra of $k[x_1, \ldots, x_n]$ and the geometry of its zero sets is the central theme of Clader–Ross.

Ideals

Definition 3.1.9 (Ideal). Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Remark (Standard quantified statement). An ideal of R is an additive subgroup of R closed under multiplication by arbitrary elements of R.

Definition 3.1.10 (Generated Ideal). Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by* $f_1, \ldots, f_m$ is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Remark (Standard quantified statement)*.* The ideal generated by $f_1, \dots, f_m$ is the set of all finite R-linear combinations of those generators.

Definition 3.1.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \dots, f_m \in R$ such that $I = \langle f_1, \dots, f_m \rangle$.

Remark (Standard quantified statement)*.* An ideal I is finitely generated when $I = \langle f_1, \dots, f_m \rangle$ for some finite list $f_1, \dots, f_m$ of elements of R.

Definition 3.1.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Remark (Standard quantified statement)*.* A commutative ring R is Noetherian when every ideal $I \subseteq R$ is finitely generated.

Theorem 3.1.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \dots, x_n]$ is Noetherian for any field k. *Go to proof.*

Remark (Standard quantified statement)*.* If R is Noetherian, then $R[x]$ is Noetherian.

Remark (Interpretation)*.* The Hilbert Basis Theorem guarantees that every ideal in $k[x_1, \dots, x_n]$ is finitely generated. This is a foundational finiteness result: it means every algebraic variety can be cut out by finitely many polynomial equations.

Quotient Rings

Definition 3.1.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Remark (Standard quantified statement)*.* For an ideal I of R, the quotient R/I is the set of additive cosets of I with operations induced from R.

Remark (Interpretation)*.* The quotient ring R/I is a commutative ring. Elements of R/I are equivalence classes under the relation $a \sim b \iff a - b \in I$.

Example (A complex quotient)*.* In $\mathbb{R}[x]$, consider the ideal $I = \langle x^2 + 1 \rangle$. The quotient ring

$$\mathbb{R}[x]/\langle x^2 + 1 \rangle \;\cong\; \mathbb{C},$$

since in the quotient, x satisfies $x^2 = -1$, which is precisely the defining relation of $i \in \mathbb{C}$.

Remark (Structural Position)*.* Quotient rings of polynomial rings are the coordinate rings of affine varieties, which are studied in depth in the next chapter. The passage

$$k[x_1, \dots, x_n] \;\longrightarrow\; k[x_1, \dots, x_n]/I$$

is the algebraic encoding of restricting from all of $\mathbb{A}^n$ to the variety $V(I)$.

Proofs

Remark (Return). Return to Theorem

Theorem (Hilbert Basis Theorem). *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k.

Professional Standard Proof.
TODO: professional standard proof for hilbert-basis-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for hilbert-basis-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

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Bibliography